

Deploying Web App on ECS - Step 4: Deploy an Application Load Balancer

To Begin with the Lab

Summary of the Lab

In this step, an **Application Load Balancer** was created to act as the public entry point for the ECS application. The load balancer was placed in the public subnet so internet users could reach the application, while the ECS tasks stayed inside private subnets for better security. A target group was created first with **IP address** as the target type so ECS tasks could register automatically later. Then the ALB was created with the correct VPC, public subnets, security group, and HTTP listener on port 80. After the load balancer was created, its DNS name was saved for use in the next ECS service step.

Understanding Application Load Balancer in ECS

An **Application Load Balancer** is the internet-facing entry point that receives user traffic and sends it to healthy ECS tasks. It exists so the application stays available and secure, because the tasks do not need public IP addresses. In this lab, the ALB sits in public subnets and forwards requests to ECS tasks running in private subnets. The benefit is simple: users can access the app from the internet, while the actual containers stay hidden and protected.

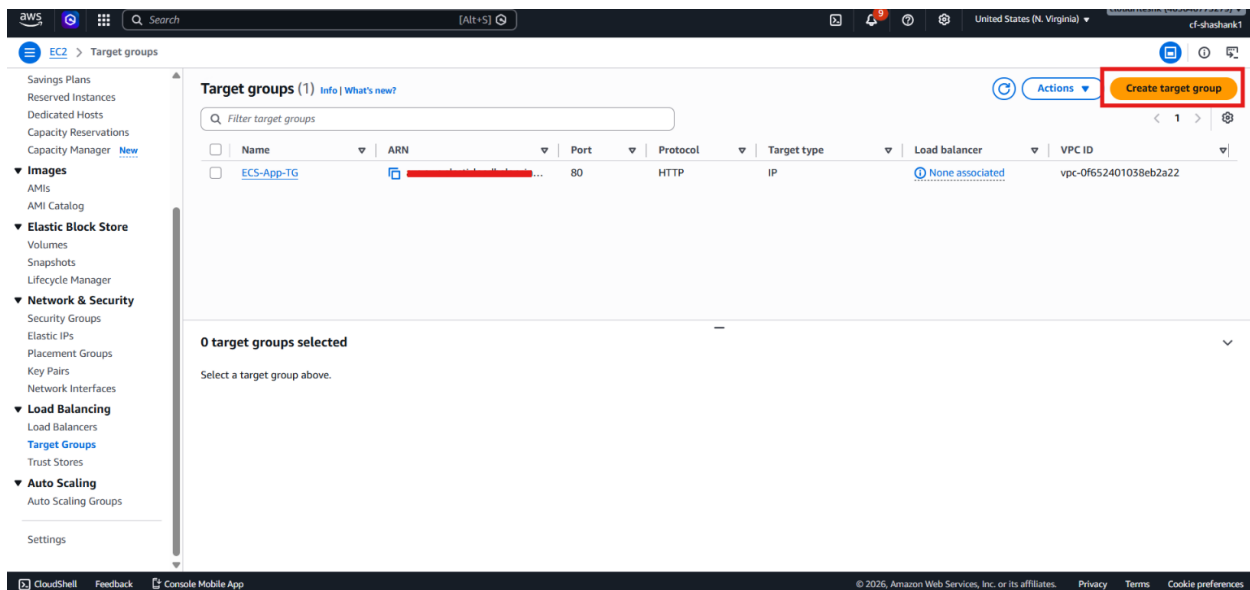
Example:

A user opens the PHP web app in a browser. The request reaches the **Application Load Balancer** on port 80, and the ALB forwards it to the ECS task running in a private subnet. If one task fails, the ALB can still send traffic to another healthy task.

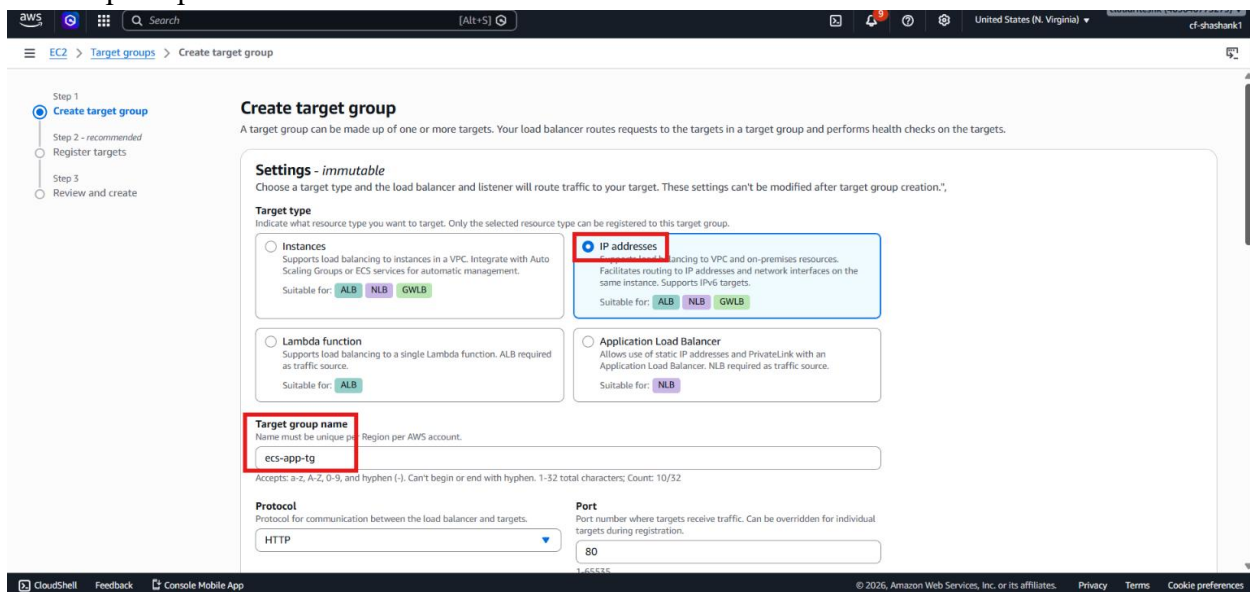
Lab Steps

Step 1 – Create the target group

- Sign in to the **AWS Management Console**.
- Open **EC2**.
- Go to **Target Groups**.
- Click **Create target group**.



- Select **IP addresses** as the target type.
- Enter the target group name: **ecs-app-tg**.
- Keep the port as 80.



- Select the correct VPC.
- Leave the remaining settings unchanged.

IP address type
Only targets with the indicated IP address type can be registered to this target group.

☒ IPv4
☐ IPv6

VPC
Select the VPC that hosts the load balancer. Only VPCs that support the IP address type selected above are available in this list. On the [Register targets](#) page, you can register IP addresses from this VPC, or from private IP addresses located outside of this load balancer's VPC (such as a peered VPC, EC2-Classic, or on-premises targets that are reachable over Direct Connect or VPN).

vpc-06c203f6d6b025c5b (app-vpc) 192.168.0.0/24 [Create VPC](#)

Protocol version

☒ HTTP1
Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2
Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC
Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks
The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol
HTTP

Health check path
Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.
/
Up to 1024 characters allowed.

- Click **Create target group**.

Step 2 – Create the Application Load Balancer

- Open EC2.
- Go to **Load Balancers**.
- Click **Create load balancer**.

Load balancers *What's new?*

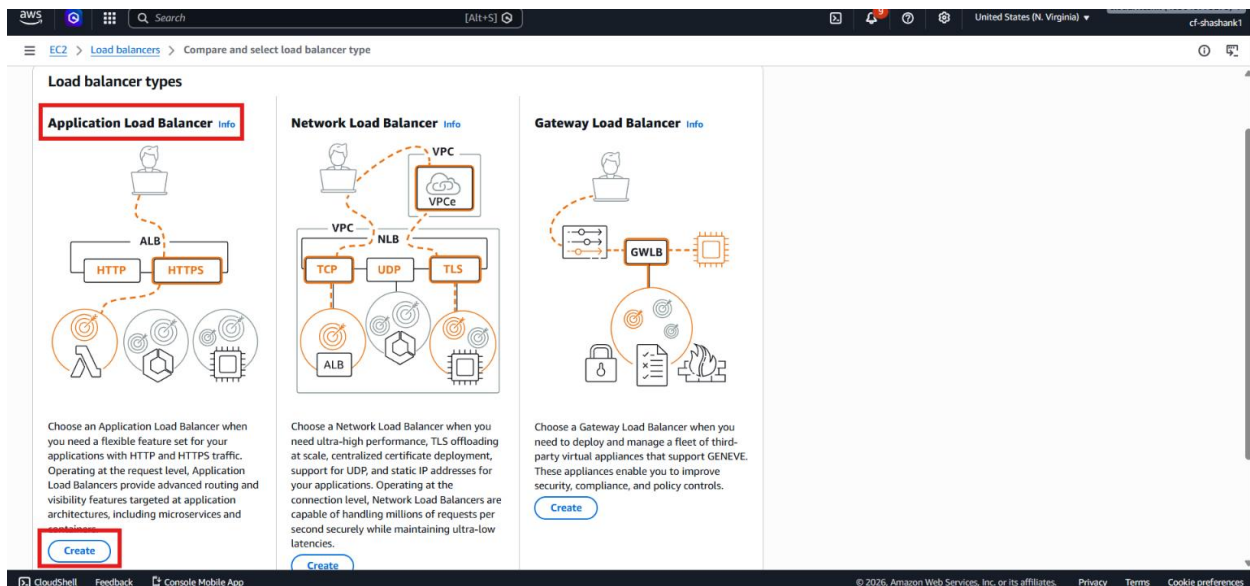
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

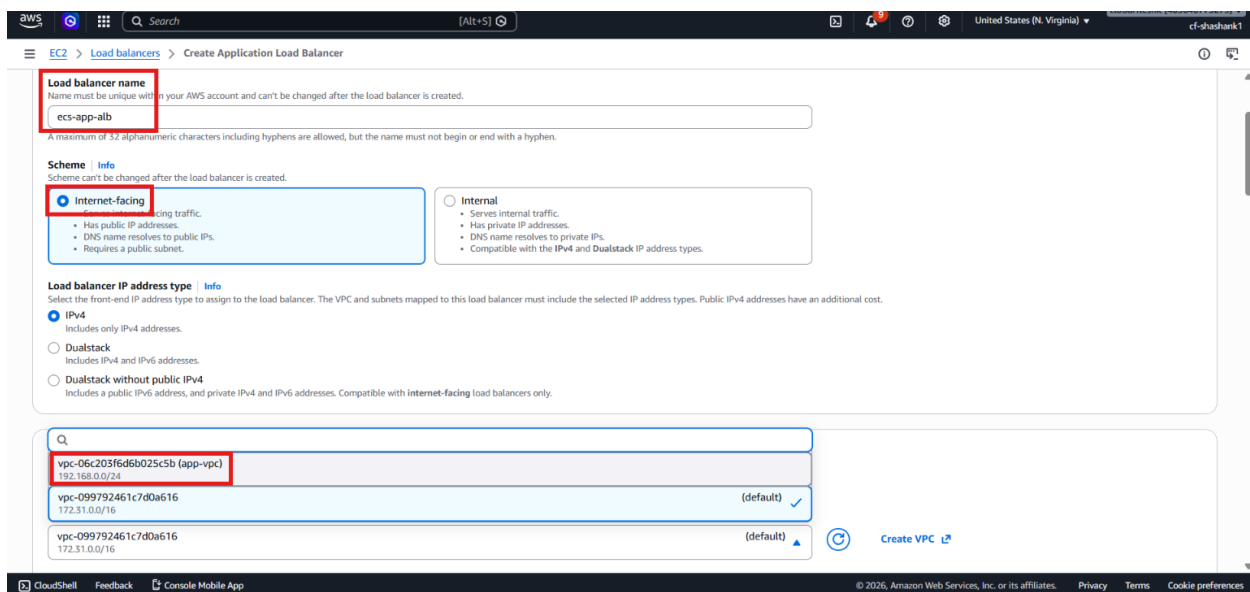
Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones	Security group
No load balancers You don't have any load balancers in us-east-1 Create load balancer							

0 load balancers selected
Select a load balancer above.

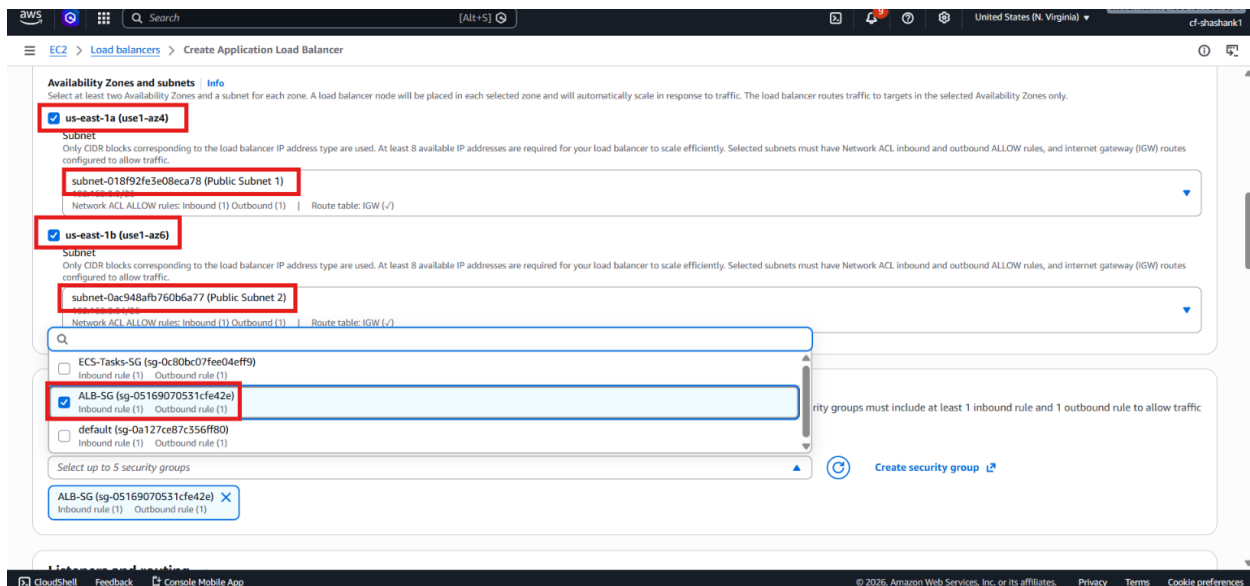
- Choose **Application Load Balancer**.



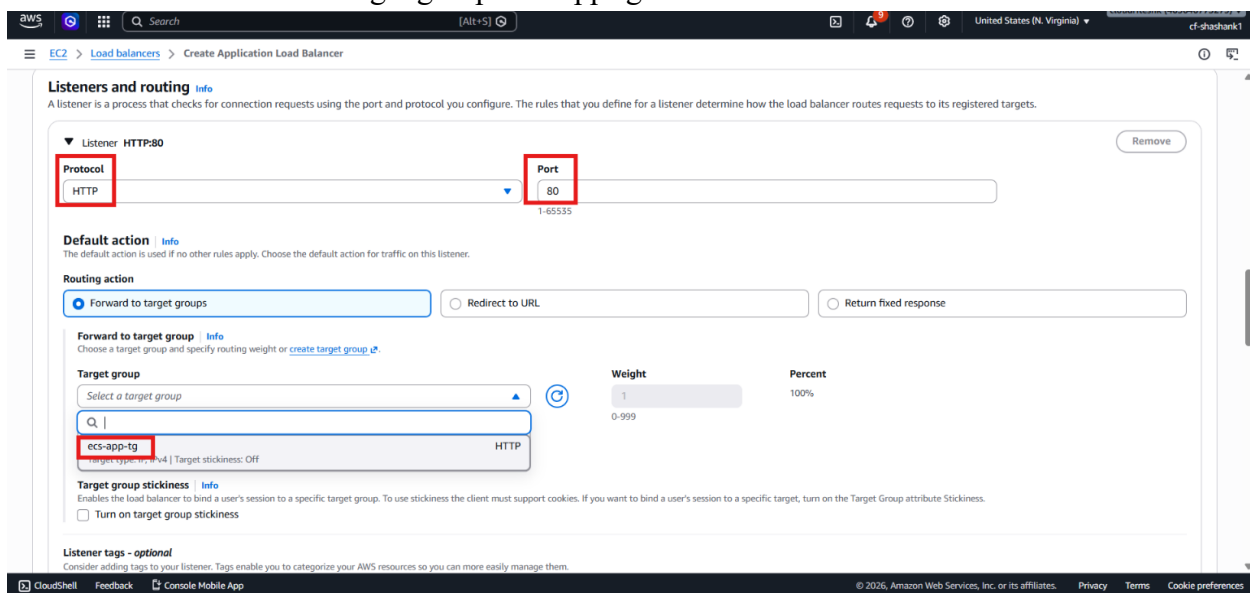
- Enter the name: **ecs-app-alb**.
- Select **Internet-facing**.
- Keep the IP type as **IPv4**.
- Choose the same VPC.



- Select both public subnets.
- Remove the default security group.
- Attach the security group created for the ALB.



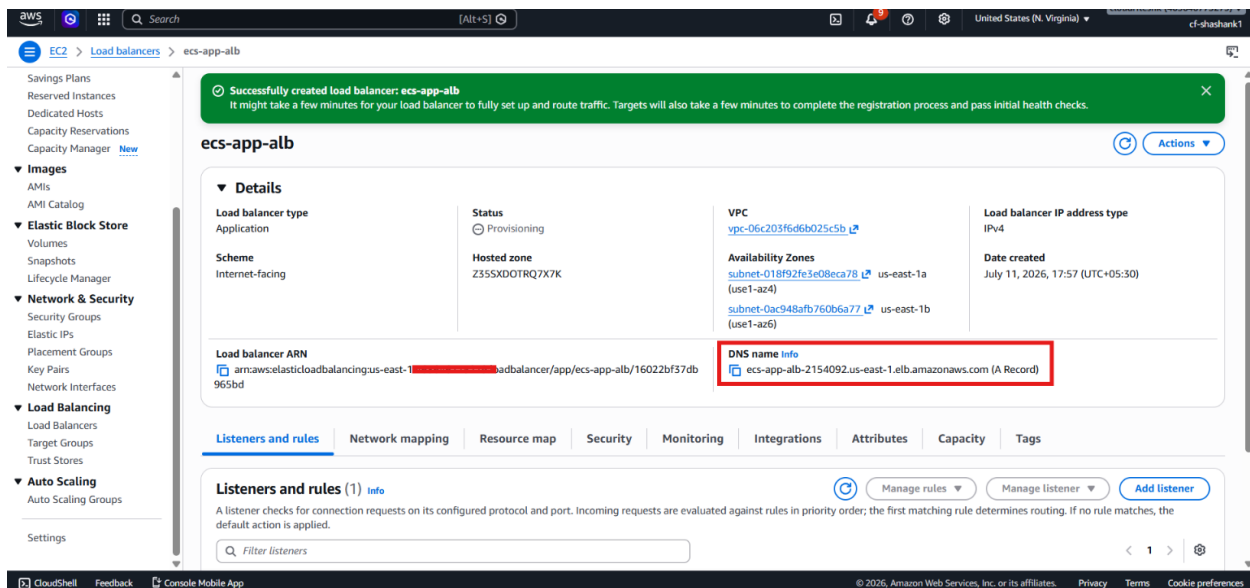
- Select the HTTP listener on port 80.
- Forward traffic to the target group: ecs-app-tg.



- Click **Create load balancer**.

Step 3 – Save the ALB DNS name

- Wait for the load balancer to finish provisioning.
- Copy the ALB DNS name.



- Save it in a note for the next step.

Step 4 – Prepare for ECS service integration

- Confirm that the ALB is in the public subnet.
- Confirm that the target group is ready for ECS task registration.
- Keep the ALB DNS name ready for the next service step.

Verification

- Verified that the target group was created successfully.
- Verified that **IP addresses** was selected as the target type.
- Verified that the **Application Load Balancer** was created successfully.
- Verified that the ALB was placed in the public subnets.
- Verified that the correct security group was attached.
- Verified that the HTTP listener on port 80 was configured.
- Verified that the ALB was attached to the correct target group.
- Verified that the ALB DNS name was saved for later use.